Exercise 1

1) $\sigma \models \Box \varphi := \neg \Diamond \neg \varphi := \neg (\top \, U \neg \varphi)$

<u>iff</u> there does NOT exist a $j \geq 0$ such that $\sigma[j \ldots] \models \neg \varphi$ and for all $0 \leq i \leq j$ we have $\sigma[i \ldots] \models \top$

<u>iff</u> there does NOT exist a $j \geq 0$ such that $\sigma[j \ldots] \models \neg \varphi$

<u>iff</u> there does NOT exist a $j \geq 0$ such that $\sigma[j \ldots] \not\models \varphi$

<u>iff</u> for all $j \geq 0$, we have

$\sigma[j \ldots] \models \varphi$

2) $\sigma \models \Box \Diamond \varphi$

<u>iff</u> for all $i \geq 0$, we have $\sigma[i \ldots] \models \Diamond \varphi$

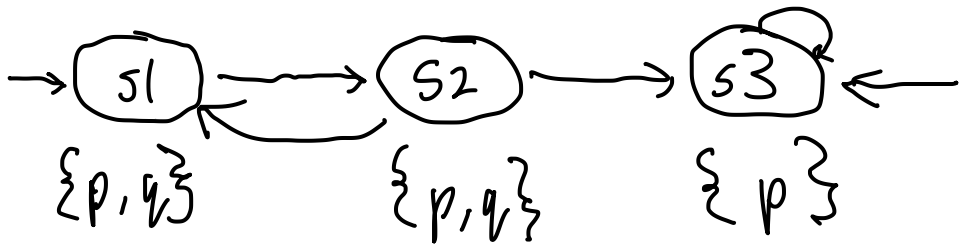
<u>iff</u> there exists $k \geq 0$ such that

$\sigma[i+k \ldots] \models \varphi$

<u>iff</u> for all $i \geq 0$ there exists $j = i+k \geq i$

such that $\sigma[j \ldots] \models \varphi$

Exercise 2



$\mathcal{T} = (S, \Phi, \rightarrow, I, A, \lambda)$:

$S = \{s1, s2, s3\}$

$\Phi = \emptyset$

$\rightarrow$ = see above

$I = \{s1, s3\}$

$A = \{p, q\}$

$\lambda = \{p\}, \{p, q\}, \{p\}$

$\sigma = s_1 s_2 s_1 s_2 \ldots \quad \{p,q\}, \{p,q\}, \ldots$

$s_1 s_2 \ldots s_3 s_3 \ldots \quad \{p,q\}, \ldots \{p\}, \ldots$

$s_3 s_3 \ldots \quad \{p\}, \{p\}, \ldots$

1) $\Box p$

$\mathcal{T} \models \Box p$ iff $s_0 \models \Box p$ (all initial states)

iff $s_1 \models \Box p$ and $s_3 \models \Box p$

iff all ρ starting at s_1 or $s_3 \models \Box p$

iff $trace(\rho) \models \Box p$

iff $\lambda(s_1) \ldots \models \Box p$ and $\lambda(s_3) \ldots \models \Box p$

All possible states contain atomic proposition p, therefore no matter the infinite path fragment ρ, $p \in \lambda(s_i)$ where $i \in \{1,2,3\}$ meaning

$\text{trace}(\rho) \models \Box p \Rightarrow \boxed{\tau \models \Box p}$

2) $\bigcirc(p \wedge q)$

$\tau \models \bigcirc(p \wedge q)$ iff s_1 and $s_2 \models \bigcirc(p \wedge q)$

iff $\sigma_1 \in S \mid s_1 \to \sigma_1$ or $s_3 \to \sigma_1 \models (p \wedge q)$

$\Rightarrow \sigma_1 \in \{s_2, s_3\}$

iff $s_2 \models p \wedge q$ and $\cancel{s_3 \models p \wedge q}$

$\{p,q\} \models p \wedge q$ and $\cancel{\{p\} \models p \wedge q}$

$\boxed{\tau \not\models \bigcirc(p \wedge q)}$

Counter example: $\sigma \approx s_3 s_3 s_3 \ldots$

$\{p\}\{p\}\{p\}\ldots$

3) $\Box(\neg q \to \Box(p \wedge \neg q))$

Essentially, this is stating that it is always the case that a state $\not\models q$ transitions to a state $\models \Box(p \wedge \neg q)$

$\Box(s_i \in S \mid s_i \not\models q \to s_j \in S \mid s_j \models \Box(p \wedge \neg q))$ for all sequences

$\{s_3\}$ $\{s_3\}$

$\Box(s3 \rightarrow s3)$

Since this is the only transition defined for s3, this holds.

Therefore, $\boxed{\mathcal{T} \models \Box(\neg q \rightarrow \Box(p \wedge \neg q))}$

4) $q \, U \, (p \wedge \neg q)$

iff s_1 and $s_3 \models q \, U \, (p \wedge \neg q)$

$s_3 \not\models q \Rightarrow s_3 \not\models q \, U \, (p \wedge \neg q)$

$\Rightarrow \boxed{\mathcal{T} \not\models q \, U \, (p \wedge \neg q)}$